

The WebXR Of Things

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Introduction

- The WebXR Of Things introduces the idea that WebXR can be the default interface to data, people, places, or services
- We explore a hyperlocal approach to looking at the world in our immediate vicinity through a generated XR experience
- Like “The Internet of Things” before it, data and services are locked up in silos.
- URLs are too long to remember, too complicated to type in XR, AR tags don’t work in XR, and so begins the woeful journey of the XR user
- The user journey to get into a WebXR experience is a challenge of its own, and we explore an open source approach, privacy friendly and distributed, to easing the entry and exit of XR experience

Methods

- PoC as a tool to explore the “dataspace”+“joincode” concept
- Devices: Meta Quest 3s, Snaap Spectacles ‘24 + a collection of IoT gadgets (various ESP32s) found in each Use Case
- A joincode server manages access to dataspaces
- A JSON schema defines the UX, not an app, and every device and service is represented by a metadata specification

Figure 1-UC1: Human Vitals

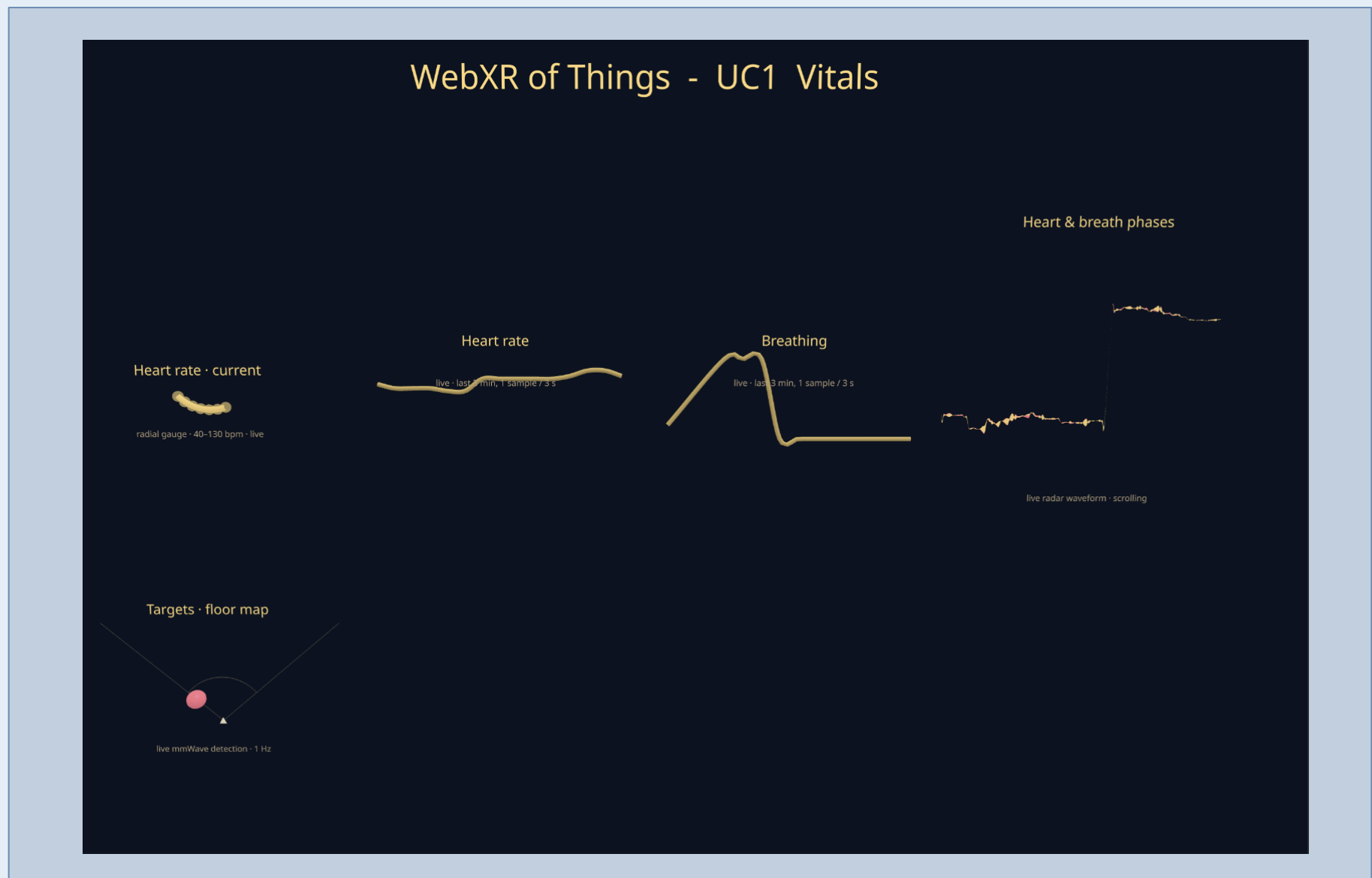


Figure 1. HR, Breathing, and “human radar”

Results

- It is possible to simplify the user journey of figuring out how to get into an XR experience by using a joincode
- Dataspaces offer a unique environment outside of a silo ecosystem, and let you explore **people** (my data, my health, who is here), **places** (a museum, this room, outside my house), and **things** (airplane entertainment system, a movie, the status of the hvac), including **services** in each
- The WebXR UX can be generated from schemas written in JSON or YAML instead of requiring a native app
- WebXR has a degraded browser experience, this must be addressed by the community, not by FANG companies
- Core Software OSS contributions: dataspace PoC, joincode server, Universal Device Metadata, Universal Service Metadata, Spatial D3.js (for ThreeJS)
- ESP32 gadgets using ESPIDFORTH (a FORTH port to ESP-IDF)
- A PoC is available on Github:
<https://github.com/IoTone/ProjectMagNET/tree/dev/reference-designs/webxroftthings/prototype/d3-spatial/demo>
- This poster, and original proposal:
<https://github.com/IoTone/ProjectMagNET/blob/dev/reference-designs/webxroftthings>

Conclusions

- The WebXR of Things can be a reality if we design it into our products
- It is possible to build in an open way on top of W3C standards to explore IoT devices, and the data around us
- Dataspaces can exist anywhere: in a home, a car, an airplane seat, at a hospital, in a museum, at this conference.
- The owner of the data should own the stack and decide their own privacy and copyright to data, not the device manufacturer or HMD maker. To force the user into adversarial outcomes of data collected by the HMD is a risk to the future of XR devices becoming mainstream
- Open Questions: how can the XR community unite to solve serious challenges with substandard WebXR browser experience? Who among us will build the first Open Hardware spatial XR glasses?

Objectives

- WebXR should be the default experience to devices, places, things
- Establish the concept of a dataspace, a hyperlocal location encapsulated in a known “joincode” to enter a particular XR experience
- Design a mechanism to generate a UX from a schema, based on device and service definitions
- Provide a reference implementation for the design, prototyped on two existing XR capable platforms
- Call out the need for more unification to solve WebXR browser issues to get feature parity with native XR experiences

Future Work

- We plan to port the demo running on a laptop to a MediaTek Genio MTK720 from GTI to act as a gateway that can be deployed into any location, including a cyberdeck
- We want to finish Spatial D3.js to get the various design primitives ported spatially
- We would like to find collaborators to build a true Open Hardware XR Glasses design

Figure 3-UC3: XRt Gallery

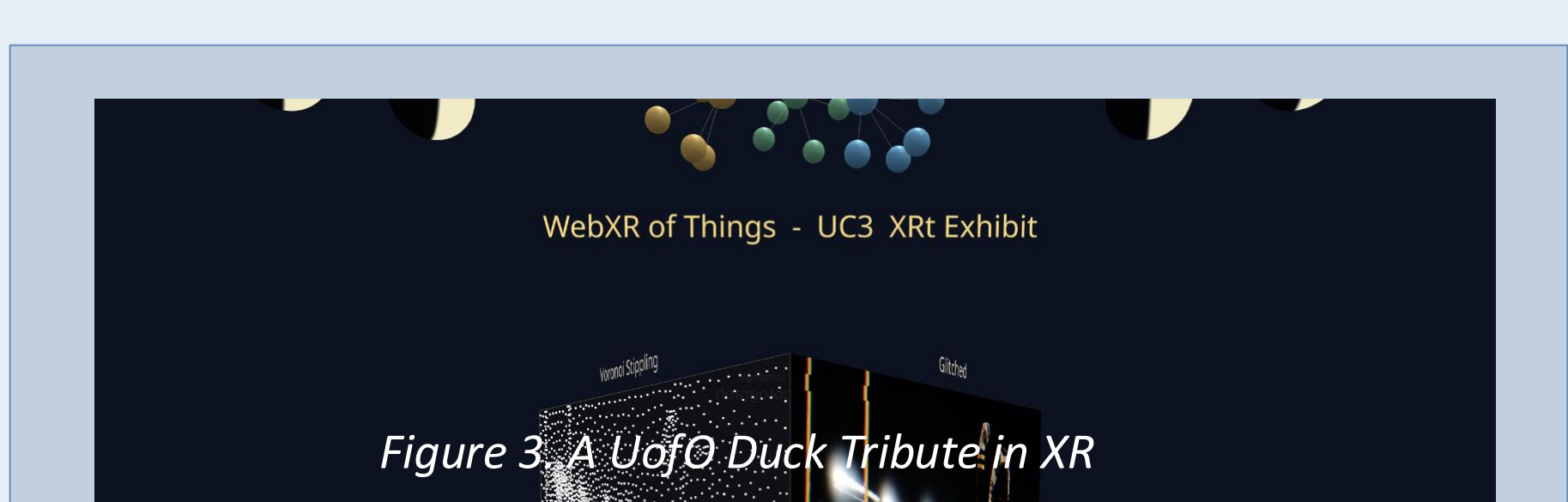


Figure 4-UC4: Airplane Entertainment System

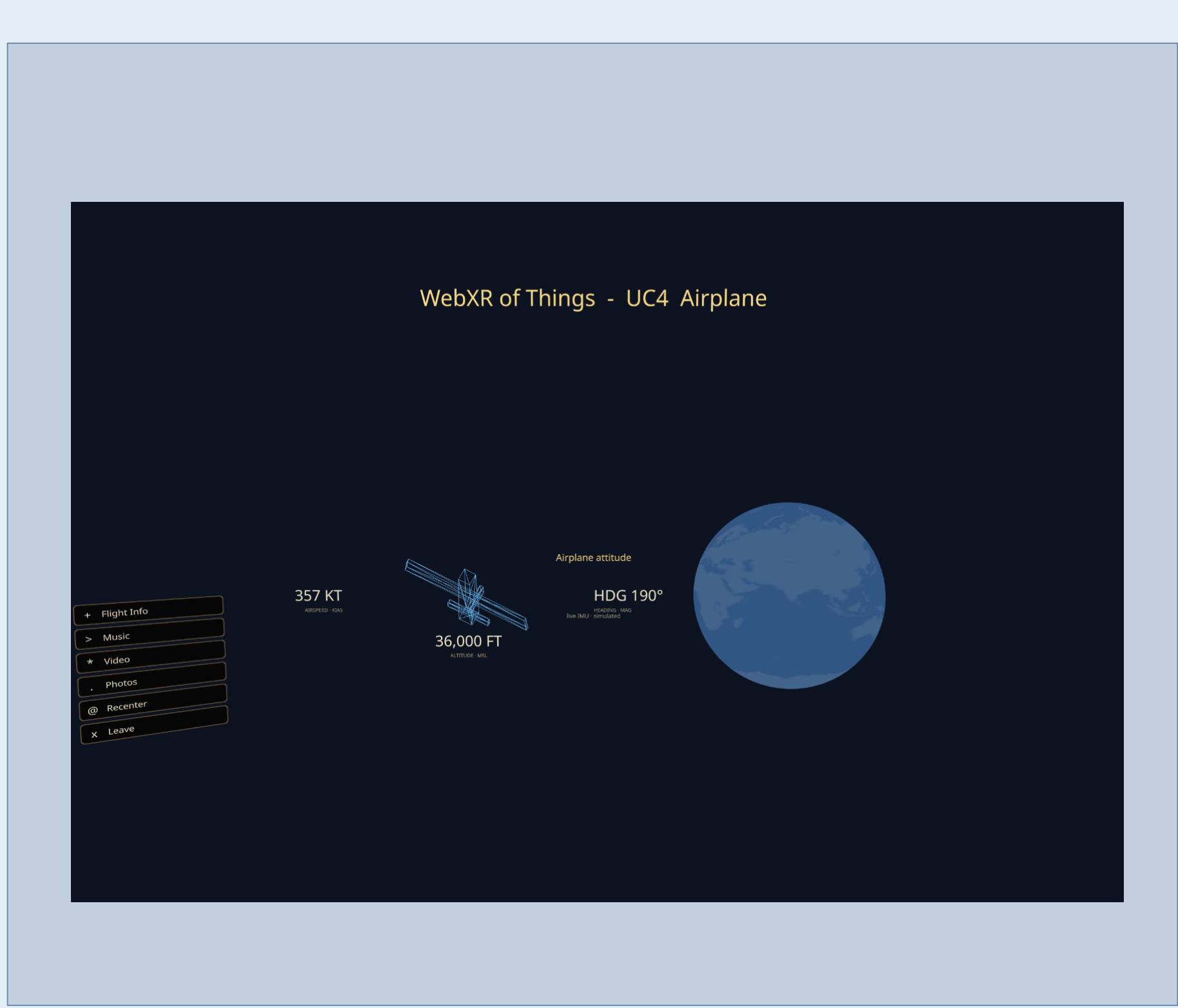
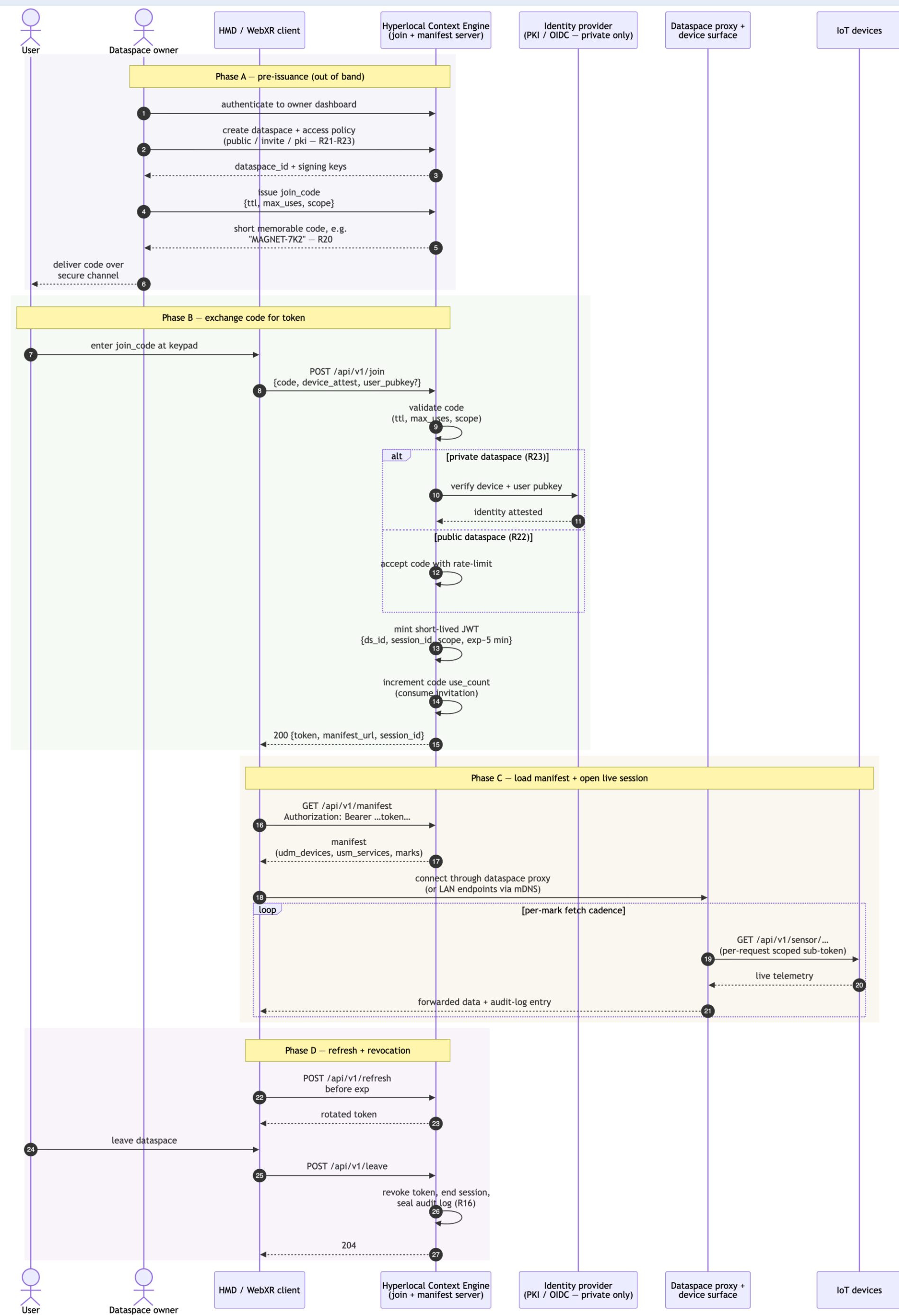


Figure 4. In-Seat Entertainment Experience: aircraft telemetry, video, audio, spatial photos



Acknowledgements

- Special thanks to IoTone, Inc (USA) for supporting the OSS work. Claude code for PoC.
- Work like this cannot happen without projects like ThreeJS, W3C standards, Chromium
- Hackathons like VisionDevCamp and MIT Reality Hack which inspire us to think spatially
- Tron (1982), the one that taught us about the data worlds locked away inside of computers
- Projects like Mozilla Hubs[1], HomeAssistant[2] for some solid ideas

References

- [1] Mozilla Hubs link codes (2020) example: <https://blog.mozilla.org/wp-content/blogs.dir/278/files/2020/10/hubs-share.png>
- [2] HomeAssistant (2013-Present): <https://www.home-assistant.io/>

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